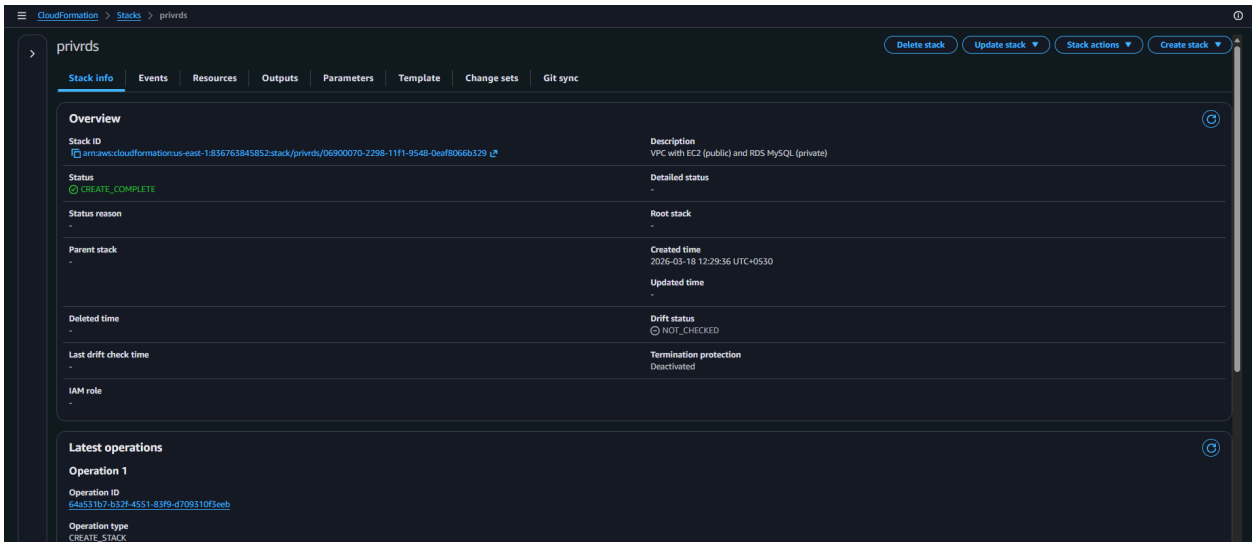
Private RDS Access

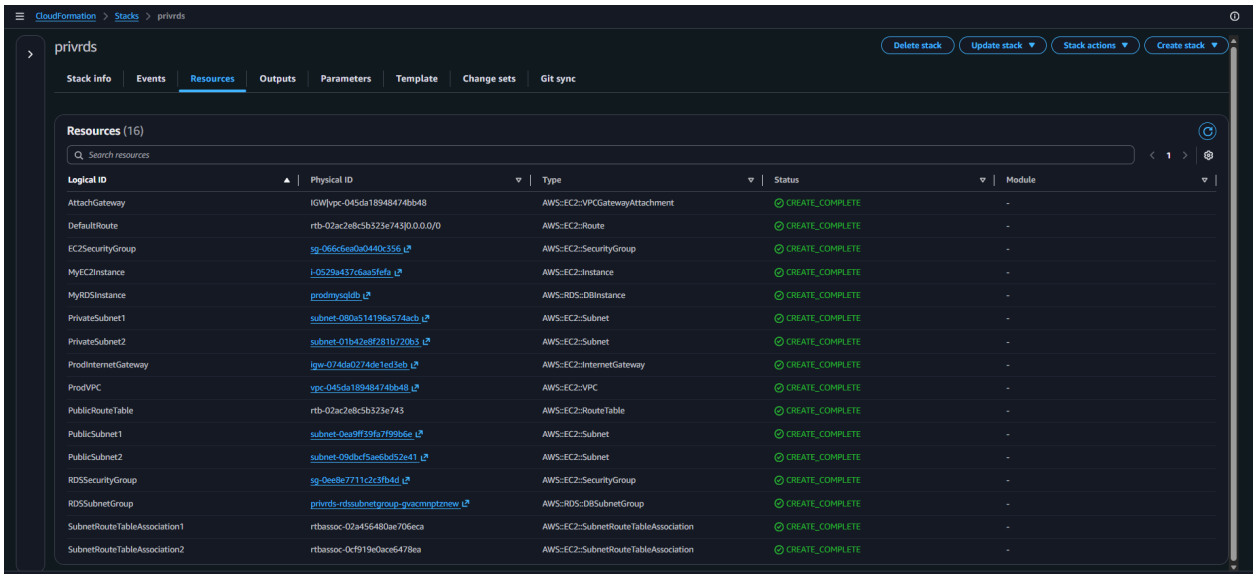
Output

```
PS E:\> ssh -i "Pub1.pem" ec2-user@ec2-13-223-96-98.compute-1.amazonaws.com
The authenticity of host 'ec2-13-223-96-98.compute-1.amazonaws.com (13.223.96.98)' can't be established.
ED25519 key fingerprint is SHA256:9Z4iCb/GT2jijFlp6C4Eevv1rGhlynU2f8yqrMH9g9c.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'ec2-13-223-96-98.compute-1.amazonaws.com' (ED25519) to the list of known hosts.
       #_
       ~\_  ####_        Amazon Linux 2023
      ~~  \_#####\
      ~~     \###|
      ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
       ~~       V~' '->
        ~~~         /
          ~~._.   _/
             _/ _/
           _/m/'
[ec2-user@ip-192-168-1-172 ~]$ yum install -y mariadb105
Error: This command has to be run with superuser privileges (under the root user on most systems).
[ec2-user@ip-192-168-1-172 ~]$ sudo yum install -y mariadb105
Last metadata expiration check: 0:07:54 ago on Wed Mar 18 07:00:33 2026.
Package mariadb105-3:10.5.29-1.amzn2023.0.1.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
[ec2-user@ip-192-168-1-172 ~]$ mysql -h prodmysqldb.cwz64mkeyewt.us-east-1.rds.amazonaws.com -u admin -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MySQL connection id is 27
Server version: 8.0.44 Source distribution

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MySQL [(none)]> USE mydatabase
ERROR 1049 (42000): Unknown database 'mydatabase'
MySQL [(none)]> CREATE DATABASE myappdb;
Query OK, 1 row affected (0.014 sec)

MySQL [(none)]>
MySQL [(none)]> SHOW DATABASES;
+--------------------+
| Database           |
+--------------------+
| information_schema |
| myappdb            |
| mysql              |
| performance_schema |
| sys                |
+--------------------+
5 rows in set (0.006 sec)

MySQL [(none)]> CREATE TABLE users (
    ->     id INT AUTO_INCREMENT PRIMARY KEY,
    ->     username VARCHAR(50) NOT NULL,
    ->     email VARCHAR(100) UNIQUE NOT NULL,
    ->     created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
    -> );
ERROR 1046 (3D000): No database selected
MySQL [(none)]> USE myappdb;
Database changed
MySQL [myappdb]> CREATE TABLE users (
    ->     id INT AUTO_INCREMENT PRIMARY KEY,
    ->     username VARCHAR(50) NOT NULL,
    ->     email VARCHAR(100) UNIQUE NOT NULL,
    ->     created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
    -> );
Query OK, 0 rows affected (0.070 sec)

MySQL [myappdb]> DESCRIBE users;
+------------+--------------+------+-----+--------------------+-------------------+
| Field      | Type         | Null | Key | Default            | Extra             |
+------------+--------------+------+-----+--------------------+-------------------+
| id         | int          | NO   | PRI | NULL               | auto_increment    |
| username   | varchar(50)  | NO   |     | NULL               |                   |
| email      | varchar(100) | NO   | UNI | NULL               |                   |
| created_at | timestamp    | YES  |     | CURRENT_TIMESTAMP  | DEFAULT_GENERATED |
+------------+--------------+------+-----+--------------------+-------------------+
4 rows in set (0.007 sec)

MySQL [myappdb]>
```

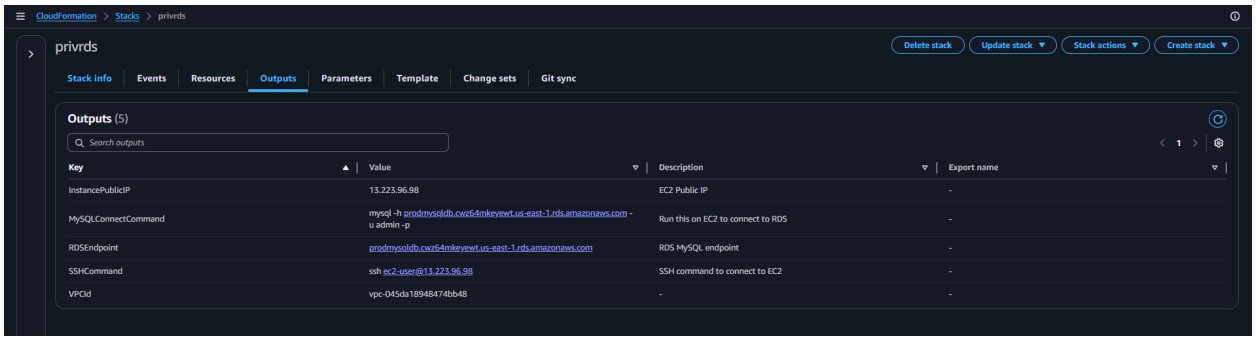
Stack Info



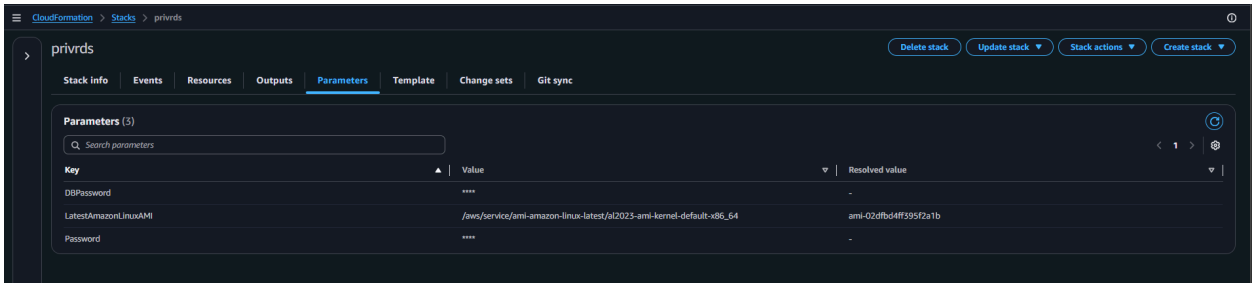
Resources



Outputs



Parameters



Template

```
{
  "AWSTemplateFormatVersion": "2010-09-09",
  "Description": "VPC with EC2 (public) and RDS MySQL (private)",
  "Parameters": {
    "LatestAmazonLinuxAMI": {
      "Type": "AWS::SSM::Parameter::Value<AWS::EC2::Image::Id>",
      "Default": "/aws/service/ami-amazon-linux-latest/al2023-ami-kernel-default-x86_64"
    },
    "Password": {
      "Type": "String",
      "NoEcho": true,
      "Description": "Password for ec2-user",
      "MinLength": 8,
      "Default": "demo12345"
    },
    "DBPassword": {
```

```
    "Type": "String",
    "NoEcho": true,
    "Description": "MySQL root password",
    "MinLength": 8,
    "Default": "demo12345"
  }
},
"Resources": {
  "ProdVPC": {
    "Type": "AWS::EC2::VPC",
    "Properties": {
      "CidrBlock": "192.168.0.0/16",
      "EnableDnsSupport": true,
      "EnableDnsHostnames": true,
      "Tags": [
        {
          "Key": "Name",
          "Value": "ProdVPC"
        }
      ]
    }
  },
  "ProdInternetGateway": {
    "Type": "AWS::EC2::InternetGateway",
    "Properties": {
      "Tags": [
        {
          "Key": "Name",
          "Value": "ProdIGW"
        }
      ]
    }
  },
  "AttachGateway": {
    "Type": "AWS::EC2::VPCGatewayAttachment",
    "Properties": {
      "VpcId": {
        "Ref": "ProdVPC"
      },
      "InternetGatewayId": {
        "Ref": "ProdInternetGateway"
      }
    }
  },
```

```
"PublicSubnet1": {
  "Type": "AWS::EC2::Subnet",
  "Properties": {
    "VpcId": {
      "Ref": "ProdVPC"
    },
    "CidrBlock": "192.168.1.0/24",
    "MapPublicIpOnLaunch": true,
    "AvailabilityZone": {
      "Fn::Select": [
        0,
        {
          "Fn::GetAZs": ""
        }
      ]
    },
    "Tags": [
      {
        "Key": "Name",
        "Value": "ProdPublicSubnet1"
      }
    ]
  }
},
"PublicSubnet2": {
  "Type": "AWS::EC2::Subnet",
  "Properties": {
    "VpcId": {
      "Ref": "ProdVPC"
    },
    "CidrBlock": "192.168.2.0/24",
    "MapPublicIpOnLaunch": true,
    "AvailabilityZone": {
      "Fn::Select": [
        1,
        {
          "Fn::GetAZs": ""
        }
      ]
    },
    "Tags": [
      {
        "Key": "Name",
        "Value": "ProdPublicSubnet2"
```

```
      }
    ]
  }
},
"PrivateSubnet1": {
  "Type": "AWS::EC2::Subnet",
  "Properties": {
    "VpcId": {
      "Ref": "ProdVPC"
    },
    "CidrBlock": "192.168.3.0/24",
    "MapPublicIpOnLaunch": false,
    "AvailabilityZone": {
      "Fn::Select": [
        0,
        {
          "Fn::GetAZs": ""
        }
      ]
    },
    "Tags": [
      {
        "Key": "Name",
        "Value": "ProdPrivateSubnet1"
      }
    ]
  }
},
"PrivateSubnet2": {
  "Type": "AWS::EC2::Subnet",
  "Properties": {
    "VpcId": {
      "Ref": "ProdVPC"
    },
    "CidrBlock": "192.168.4.0/24",
    "MapPublicIpOnLaunch": false,
    "AvailabilityZone": {
      "Fn::Select": [
        1,
        {
          "Fn::GetAZs": ""
        }
      ]
    },
```

```
    "Tags": [
      {
        "Key": "Name",
        "Value": "ProdPrivateSubnet2"
      }
    ]
  }
},
"PublicRouteTable": {
  "Type": "AWS::EC2::RouteTable",
  "Properties": {
    "VpcId": {
      "Ref": "ProdVPC"
    },
    "Tags": [
      {
        "Key": "Name",
        "Value": "ProdPublicRT"
      }
    ]
  }
},
"DefaultRoute": {
  "Type": "AWS::EC2::Route",
  "DependsOn": "AttachGateway",
  "Properties": {
    "RouteTableId": {
      "Ref": "PublicRouteTable"
    },
    "DestinationCidrBlock": "0.0.0.0/0",
    "GatewayId": {
      "Ref": "ProdInternetGateway"
    }
  }
},
"SubnetRouteTableAssociation1": {
  "Type": "AWS::EC2::SubnetRouteTableAssociation",
  "Properties": {
    "SubnetId": {
      "Ref": "PublicSubnet1"
    },
    "RouteTableId": {
      "Ref": "PublicRouteTable"
    }
```

```
    }
  },
  "SubnetRouteTableAssociation2": {
    "Type": "AWS::EC2::SubnetRouteTableAssociation",
    "Properties": {
      "SubnetId": {
        "Ref": "PublicSubnet2"
      },
      "RouteTableId": {
        "Ref": "PublicRouteTable"
      }
    }
  },
  "EC2SecurityGroup": {
    "Type": "AWS::EC2::SecurityGroup",
    "Properties": {
      "GroupDescription": "Allow SSH and HTTP",
      "VpcId": {
        "Ref": "ProdVPC"
      },
      "SecurityGroupIngress": [
        {
          "IpProtocol": "tcp",
          "FromPort": 22,
          "ToPort": 22,
          "CidrIp": "0.0.0.0/0"
        },
        {
          "IpProtocol": "tcp",
          "FromPort": 80,
          "ToPort": 80,
          "CidrIp": "0.0.0.0/0"
        }
      ],
      "Tags": [
        {
          "Key": "Name",
          "Value": "EC2SG"
        }
      ]
    }
  },
  "RDSSecurityGroup": {
    "Type": "AWS::EC2::SecurityGroup",
```

```
  "Properties": {
    "GroupDescription": "Allow MySQL only from EC2 security group",
    "VpcId": {
      "Ref": "ProdVPC"
    },
    "SecurityGroupIngress": [
      {
        "IpProtocol": "tcp",
        "FromPort": 3306,
        "ToPort": 3306,
        "SourceSecurityGroupId": {
          "Ref": "EC2SecurityGroup"
        }
      }
    ],
    "Tags": [
      {
        "Key": "Name",
        "Value": "RDSSG"
      }
    ]
  }
},
"RDSSubnetGroup": {
  "Type": "AWS::RDS::DBSubnetGroup",
  "Properties": {
    "DBSubnetGroupDescription": "Private subnets for RDS",
    "SubnetIds": [
      {
        "Ref": "PrivateSubnet1"
      },
      {
        "Ref": "PrivateSubnet2"
      }
    ]
  }
},
"MyRDSInstance": {
  "Type": "AWS::RDS::DBInstance",
  "Properties": {
    "DBInstanceIdentifier": "prodmysqldb",
    "DBInstanceClass": "db.t3.micro",
    "Engine": "mysql",
    "EngineVersion": "8.0",
```

```
      "MasterUsername": "admin",
      "MasterUserPassword": {
        "Ref": "DBPassword"
      },
      "AllocatedStorage": 20,
      "StorageType": "gp2",
      "PubliclyAccessible": false,
      "MultiAZ": false,
      "DBSubnetGroupName": {
        "Ref": "RDSSubnetGroup"
      },
      "VPCSecurityGroups": [
        {
          "Ref": "RDSSecurityGroup"
        }
      ],
      "DeletionProtection": false
    }
  },
  "MyEC2Instance": {
    "Type": "AWS::EC2::Instance",
    "Properties": {
      "InstanceType": "t3.micro",
      "ImageId": {
        "Ref": "LatestAmazonLinuxAMI"
      },
      "SubnetId": {
        "Ref": "PublicSubnet1"
      },
      "SecurityGroupIds": [
        {
          "Ref": "EC2SecurityGroup"
        }
      ],
      "KeyName": "Pub1",
      "UserData": {
        "Fn::Base64": {
          "Fn::Sub": "#!/bin/bash\ndnf update -y\ndnf install -y nginx\nsystemctl start
nginx\nsystemctl enable nginx\necho \"ec2-user:${Password}\" | chpasswd\nsed -i
's/PasswordAuthentication no/PasswordAuthentication yes/g' /etc/ssh/sshd_config\nsed -i
's/#PasswordAuthentication yes/PasswordAuthentication yes/g' /etc/ssh/sshd_config\nsed -i
's/PermitRootLogin no/PermitRootLogin yes/g' /etc/ssh/sshd_config\nsystemctl restart sshd\ndnf
install -y mariadb105\n"
        }
```

```
        }
      }
    }
  },
  "Outputs": {
    "InstancePublicIP": {
      "Description": "EC2 Public IP",
      "Value": {
        "Fn::GetAtt": [
          "MyEC2Instance",
          "PublicIp"
        ]
      }
    },
    "SSHCommand": {
      "Description": "SSH command to connect to EC2",
      "Value": {
        "Fn::Sub": "ssh ec2-user@${MyEC2Instance.PublicIp}"
      }
    },
    "RDSEndpoint": {
      "Description": "RDS MySQL endpoint",
      "Value": {
        "Fn::GetAtt": [
          "MyRDSInstance",
          "Endpoint.Address"
        ]
      }
    },
    "MySQLConnectCommand": {
      "Description": "Run this on EC2 to connect to RDS",
      "Value": {
        "Fn::Sub": "mysql -h ${MyRDSInstance.Endpoint.Address} -u admin -p"
      }
    },
    "VPCId": {
      "Value": {
        "Ref": "ProdVPC"
      }
    }
  }
}
```